Deep Learning Based Channel Estimation for 5G/6G Wireless Systems

Phase I: Reproduction of AttRNet for OFDM Channel Estimation and Development of a Custom Simulation Environment

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Project Repository: https://github.com/priyanshu2393/MTP_Attnet

Abstract

Accurate channel state information (CSI) is required for reliable signal detection and equalization in orthogonal frequency-division multiplexing (OFDM) systems. Least-squares (LS) channel estimation is simple but sensitive to noise, while linear minimum mean-square-error (LMMSE) estimation relies on channel statistics and noise information.

This project studies AttRNet, an attention based residual neural network proposed for refining pilot-based LS channel estimates. The first phase focuses on reproducing the method in a conventional single-link OFDM setting using the 3GPP EPA multipath channel model.

Since the original paper does not provide a public dataset or implementation, a custom simulation and data-processing pipeline was developed using MATLAB and Python, followed by PyTorch implementations of AttRNet-FC and AttRNet-Conv.

1. Introduction and Project Scope

1.1 Background

Wireless signals propagate through environments containing buildings, terrain, objects and other scatterers. Reflection, diffraction, scattering, attenuation and multipath delay alter the transmitted waveform. The receiver must estimate the resulting wireless channel to compensate for these effects during equalization and data detection. In OFDM, the available bandwidth is divided among orthogonal subcarriers, and the channel can vary across both frequency and time.

Pilot symbols are known reference symbols inserted into the transmitted resource grid. At pilot locations, the receiver can compare the known transmitted values with the received values to obtain channel observations. LS estimation provides a simple estimate at these positions, but noise can make the observations inaccurate. Deep learning based refinement uses training examples to learn channel structure and improve estimates from noisy or sparse observations.

1.2 Project Scope and Phases

The broader project proposal targets pilot-efficient OFDM channel estimation and a possible extension to RIS-assisted cascaded channels under CDL-C and CDL-D propagation models. The current phase, however, is limited to reproducing the AttRNet method in the conventional single-link EPA-OFDM setting used as the baseline in the selected paper

1.3 Phase I Problem Statement

Reproduce the AttRNet-based OFDM channel-estimation framework described by Gao et al. by implementing its attention-residual network variants, constructing a custom EPA-OFDM dataset, and evaluating channel-estimation MSE across the reported pilot configurations and SNR conditions. The replication must document implementation assumptions where the paper does not provide sufficient detail and assess whether the reproduced trends and numerical results are consistent with the published results.

1.4 Broader Project Objective

Following completion and validation of the baseline, the broader objective is to investigate a deep learning method that reduces pilot overhead while maintaining channel-estimation accuracy, and subsequently examine extension to the project-proposal setting. The specific novel technique and final target system configuration remain to be finalized with the supervisor.

2. Baseline Paper and Technical Method

2.1 Paper Overview

The selected baseline paper, “Deep residual learning with attention mechanism for OFDM channel estimation” by W. Gao, W. Zhang, L. Liu and M. Yang, proposes AttRNet to refine noisy LS channel estimates. It presents convolutional and fully connected variants, AttRNet-Conv and AttRNet-FC. Both combine residual learning with channel attention, while differing in the implementation of the attention excitation layers.

2.2 OFDM Channel Observation

For subcarrier i and OFDM symbol j, the received signal is represented as:

$$y[i,j] = h[i,j]\, x[i,j] + w[i,j]$$

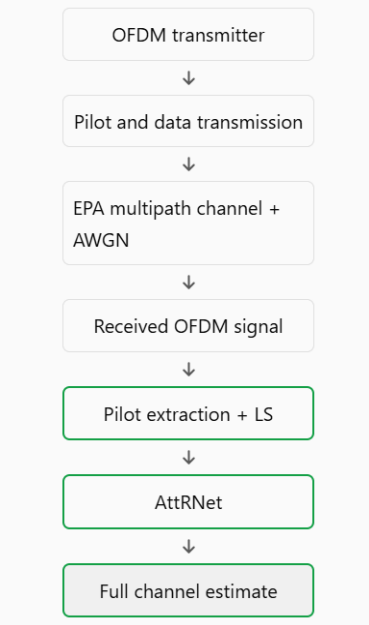
x[i,j] → transmitted symbol
h[i,j] → channel coefficient
w[i,j] → gaussian noise

At pilot locations, x is known to the receiver. The LS estimate is obtained by dividing the received pilot by its known transmitted value:

$$\hat{h}LS[i,j] = y[i,j] / x[i,j] \quad \text{(at pilot positions)}$$

Because pilots occupy only a subset of the time-frequency resource grid, the initial LS observations are sparse. AttRNet processes the real and imaginary components of these observations and produces an estimate over the full channel grid.

2.3 AttRNet: Residual Learning and Attention



AttrNet model Architecture:

Input Matrix H_{LS}

For an OFDM grid, $\mathbf{H}_{LS} \in \mathbb{C}^{N_P^Q \times N_P^M}$

where each element $H_{LS}[k, n]$

represents the estimated **complex channel gain** at a particular subcarrier k and OFDM symbol n.

Each element contains: $H[k,n] = |H[k,n]|e^{j\phi[k,n]}$

so it contains:

- **magnitude** → attenuation/amplification
- **phase** → phase shift caused by the channel

The LS estimate is obtained from the known pilots:

$$\hat{H}_{LS}[k_p, n_p] = \frac{Y[k_p, n_p]}{X[k_p, n_p]}$$

The paper then separates the complex matrix into **real and imaginary parts**, making two input channels.

$$\hat{H}_{LS} \longrightarrow \begin{bmatrix} \Re(\hat{H}_{LS}) \\ \Im(\hat{H}_{LS}) \end{bmatrix}$$

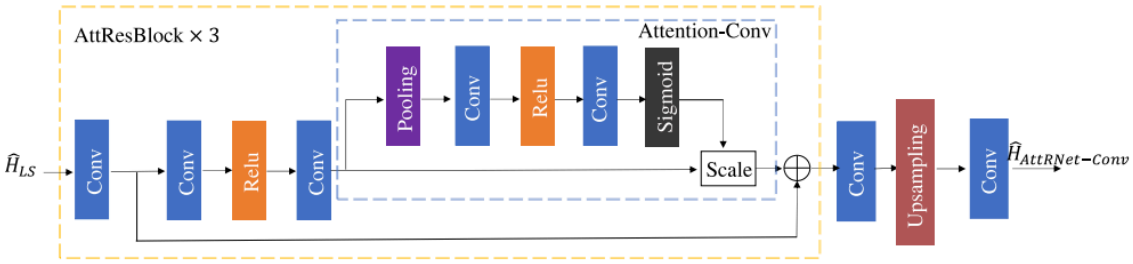


Fig. 2. Architecture of AttRNet-Conv.

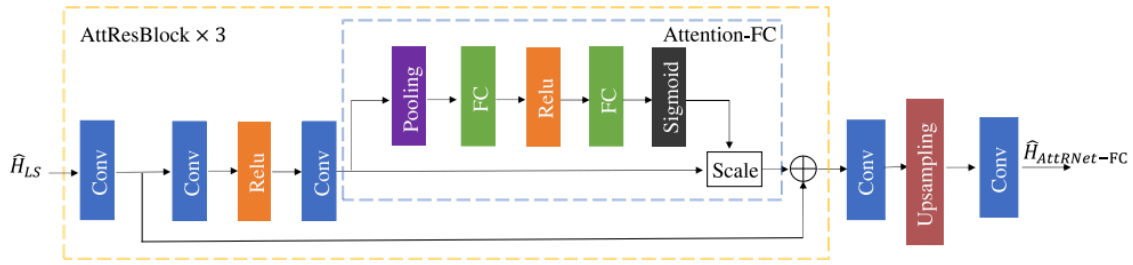


Fig. 3. Architecture of AttRNet-FC.

Inside every AttResBlock, the feature map goes through 3 convolutions and then the **attention module**.

Convolution Block:

The convolution looks at **local regions of the channel matrix** and learns spatial/time-frequency patterns.

For example, a 3×3 kernel sees:

$$\begin{bmatrix} H_{k-1,n-1} & H_{k,n-1} & H_{k+1,n-1} \\ H_{k-1,n} & H_{k,n} & H_{k+1,n} \\ H_{k-1,n+1} & H_{k,n+1} & H_{k+1,n+1} \end{bmatrix}$$

and learns useful channel structures from them.

So convolution is basically learning:

$$\boxed{\text{local time-frequency channel patterns}}$$

The input to the attention module is a **feature map produced by the third convolution**, and it contains learned **local time-frequency channel patterns/features**.

$$F \in \mathbb{R}^{H \times W \times N_K}$$

Here N_K is the number of feature channels

Attention Block:

1. Squeeze operation through Pooling

The **Pooling** in the attention block performs **global average pooling** over the spatial/time-frequency dimensions:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F_c(i,j)$$

So each feature channel F_c, which is an $H \times W$ matrix, is reduced to **one number**.

$$H \times W \times N_K \rightarrow 1 \times 1 \times N_K$$

It summarizes the **global importance/response of each feature channel** rather than keeping every spatial location. Then apply attention to these feature channels, **attention weights tell you which learned features are important**

2. Attention-FC

After pooling: $1 \times 1 \times N_K$

is passed through fully connected layers :

$$\boxed{z \rightarrow \mathrm{FC} \rightarrow \mathrm{ReLU} \rightarrow \mathrm{FC} \rightarrow \mathrm{Sigmoid}}$$

The FC layers learn the **relationship between the feature channels** and produce a weight s_c for each channel:

$$s_c \in [0, 1].$$

Then: $\boxed{F_c' = s_c F_c}$

This feature channel is important → give it a high weight.

This feature channel is less useful/noisy → give it a low weight.

3. Attention-Conv

The purpose is **the same**, but instead of FC layers it uses 1×1 convolutions:

$$\boxed{z \to 1 \times 1 \text{ Conv} \to \text{ReLU} \to 1 \times 1 \text{ Conv} \to \text{Sigmoid}}$$

These 1×1 convolutions operate across the **feature-channel dimension** and learn the channel relationships.

They again produce:

$$s_1, s_2, \ldots, s_{N_K}$$

and scale the original feature map: $\boxed{F_c' = s_c F_c}$

$$\boxed{\underbrace{\text{Conv} \to \text{Conv} \to \text{ReLU} \to \text{Conv}}_{\text{Feature extraction}} \to \underbrace{\text{Attention} - \text{FC/Conv}}_{\text{Feature weighting}}}$$

Then attention produces channel weights, the **Scale** operation multiplies those weights with the feature map, and finally the result is added through the **residual/skip connection**.

After the **AttResBlocks**, the network still has a **learned feature representation**, not yet the final 72×14 channel matrix.

The next two operations have different purposes:

1. Convolution after AttResBlocks

The 3×3 convolution combines the refined feature maps and transforms them into a representation suitable for reconstruction.

2. Upsampling

The LS input is **sparse**, e.g. for 24 pilots:

$$6 \times 4$$

but the desired channel is:

$$72 \times 14.$$

Therefore, the network needs to increase the spatial/time-frequency resolution:

AttRNet uses **bilinear upsampling** for this

Supervised Learning Model:

Y(true): clean channel target generated by simulation

Y(predicted): a noisy pilot based LS

Objective: minimize the discrepancy between the predicted channel and the target using MSE

Error Calculation:

The paper uses **MSE** as the training loss:

$$\boxed{\mathcal{L} = \frac{1}{N} \sum \left| H_{\text{true}} - \hat{H}_{\text{AttRNet}} \right|^2}$$

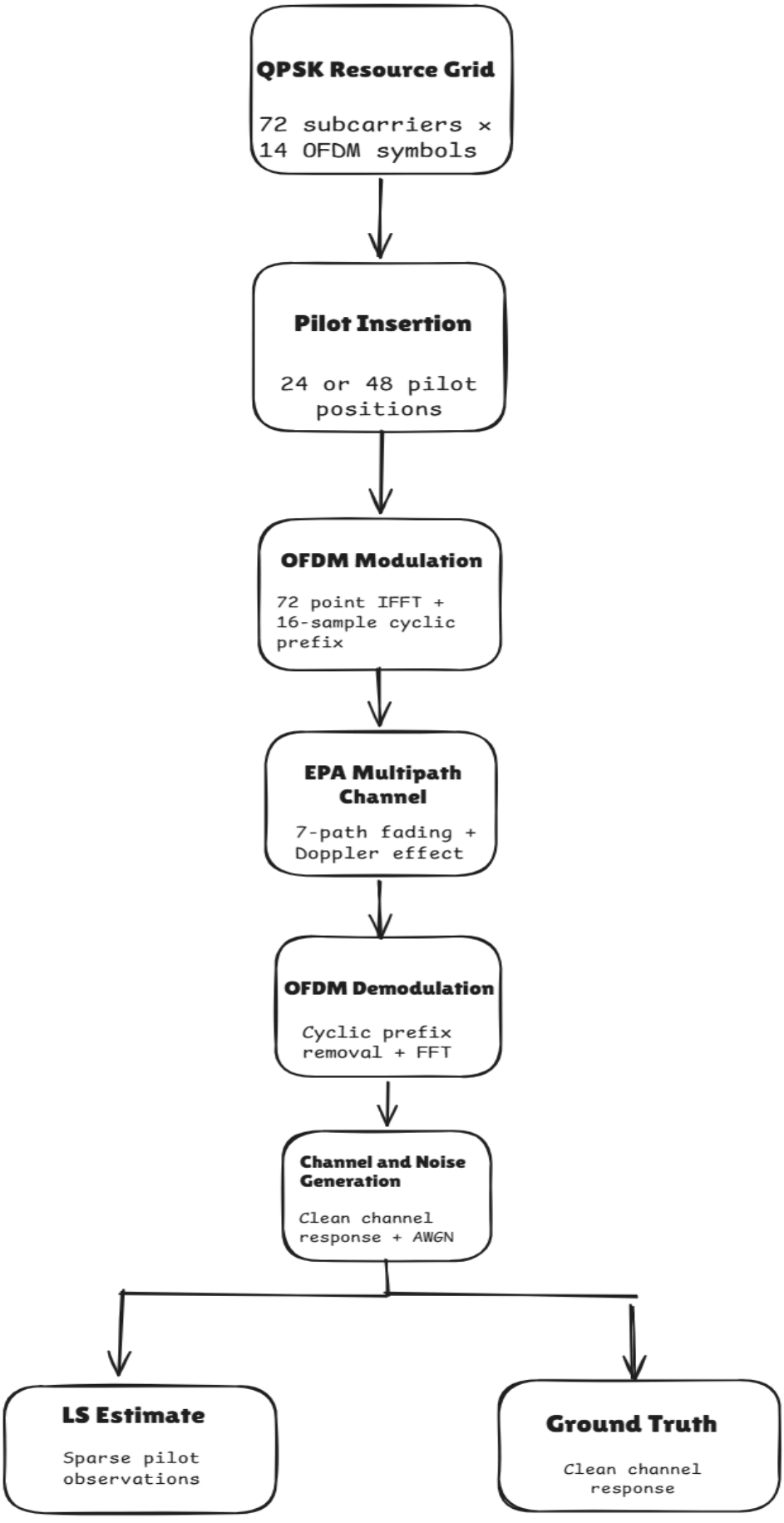
3. Simulation Environment and Dataset

3.1 Parameters taken from the paper

Parameter	Value
Propagation model	3GPP Extended Pedestrian A (EPA), 7 paths
EPA delays	0, 30, 70, 90, 110, 190, 410 ns
EPA relative powers	0, -1, -2, -3, -8, -17.2, -20.8 dB
Carrier frequency	2.5 GHz
Receiver speed	50 km/h
Subcarrier spacing	15 kHz
Active subcarriers	72
OFDM symbols per subframe	14
Modulation	QPSK
Cyclic prefix length	16 samples
Pilot-count experiments	24 and 48
Dataset size	20,000 samples
Split	12,000 train / 4,000 validation / 4,000 test

3.2 Dataset Generation

The dataset is generated using a MATLAB waveform-level OFDM simulator with the EPA multipath fading channel


QPSK Resource Grid
72 subcarriers × 14 OFDM symbols
Pilot Insertion
24 or 48 pilot positions
OFDM Modulation
72 point IFFT + 16-sample cyclic prefix
EPA Multipath Channel
7-path fading + Doppler effect
OFDM Demodulation
Cyclic prefix removal + FFT
Channel and Noise Generation
Clean channel response + AWGN
LS Estimate
Sparse pilot observations
Ground Truth
Clean channel response

4. Results

4.1 Completed Experiments

Training and evaluation have been completed for four configurations of AttRNet: AttRNet-FC and AttRNet-Conv, each evaluated using 24 and 48 pilot symbols. The models were evaluated at SNR values of 0, 5, 10, 15, and 20 dB using the custom-generated EPA-OFDM dataset.

The mean squared error (MSE) between the estimated channel and the ground-truth channel was used as the evaluation metric.

SNR (dB)	FC, 24 pilots	Conv, 24 pilots	FC, 48 pilots	Conv, 48 pilots
0	0.065501	0.065502	0.038934	0.038917
5	0.025519	0.025524	0.016112	0.016113
10	0.011482	0.011486	0.007570	0.007568
15	0.006338	0.006349	0.004428	0.004420
20	0.004498	0.004511	0.003134	0.003127

The results show a consistent decrease in channel-estimation MSE as the SNR increases. This behavior is expected because higher SNR provides less noise-corrupted channel observations, allowing more accurate estimation.

4.2 Approximate Comparison with the Paper

Table 1. AttRNet-FC with 24 Pilots

SNR (dB)	Paper MSE (approx.)	Reproduced MSE	Difference (%)
0	0.23400	0.065501	-72.01
5	0.09100	0.025519	-71.96
10	0.03160	0.011482	-63.67
15	0.01200	0.006338	-47.18
20	0.00394	0.004498	+14.16

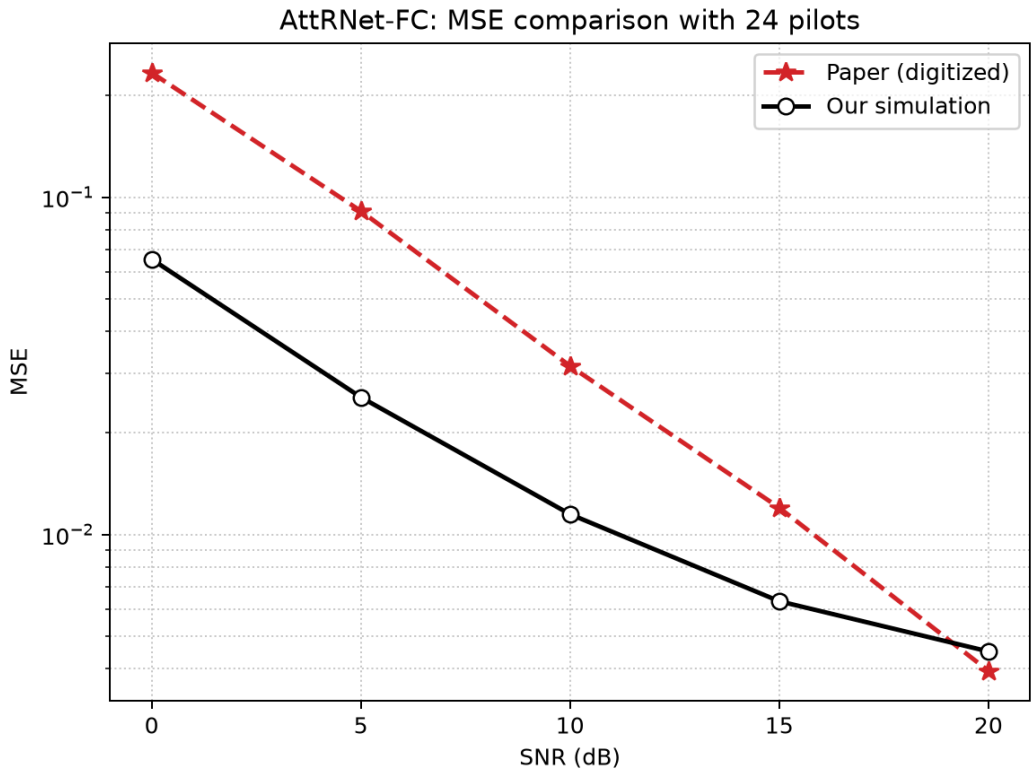


Table 2. AttRNet-FC with 48 Pilots

SNR (dB)	Paper MSE (approx.)	Reproduced MSE	Difference (%)
0	0.12800	0.038934	-69.58
5	0.05050	0.016112	-68.09
10	0.01840	0.007570	-58.86
15	0.00606	0.004428	-26.93
20	0.00194	0.003134	+61.55

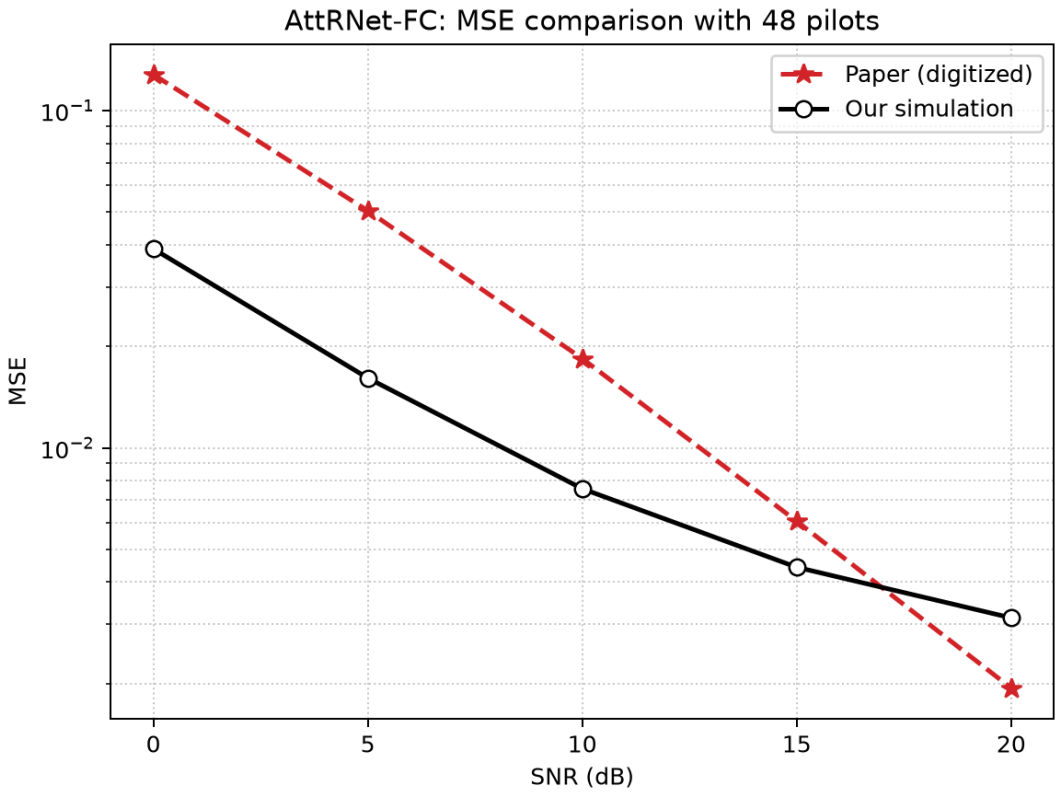


Table 3. AttRNet-Conv with 24 Pilots

SNR (dB)	Paper MSE (approx.)	Reproduced MSE	Difference (%)
0	0.21600	0.065502	-69.68
5	0.09120	0.025524	-72.01
10	0.03160	0.011486	-63.65
15	0.01200	0.006349	-47.09
20	0.00433	0.004511	+4.18

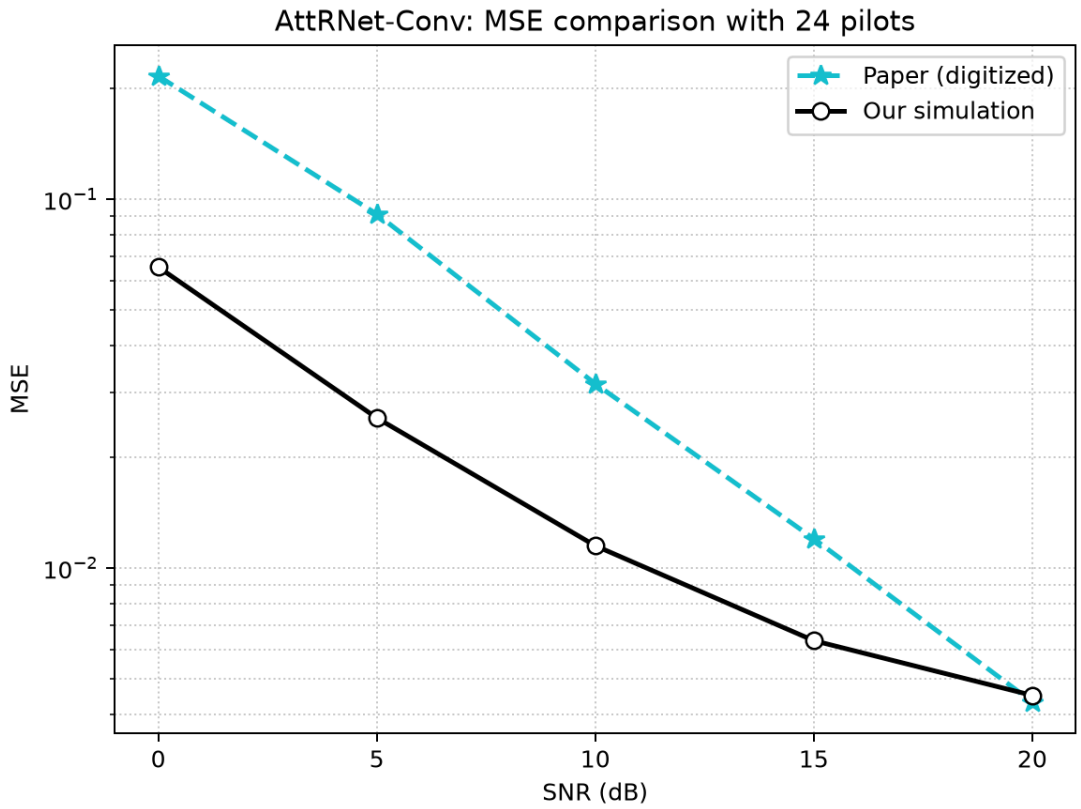
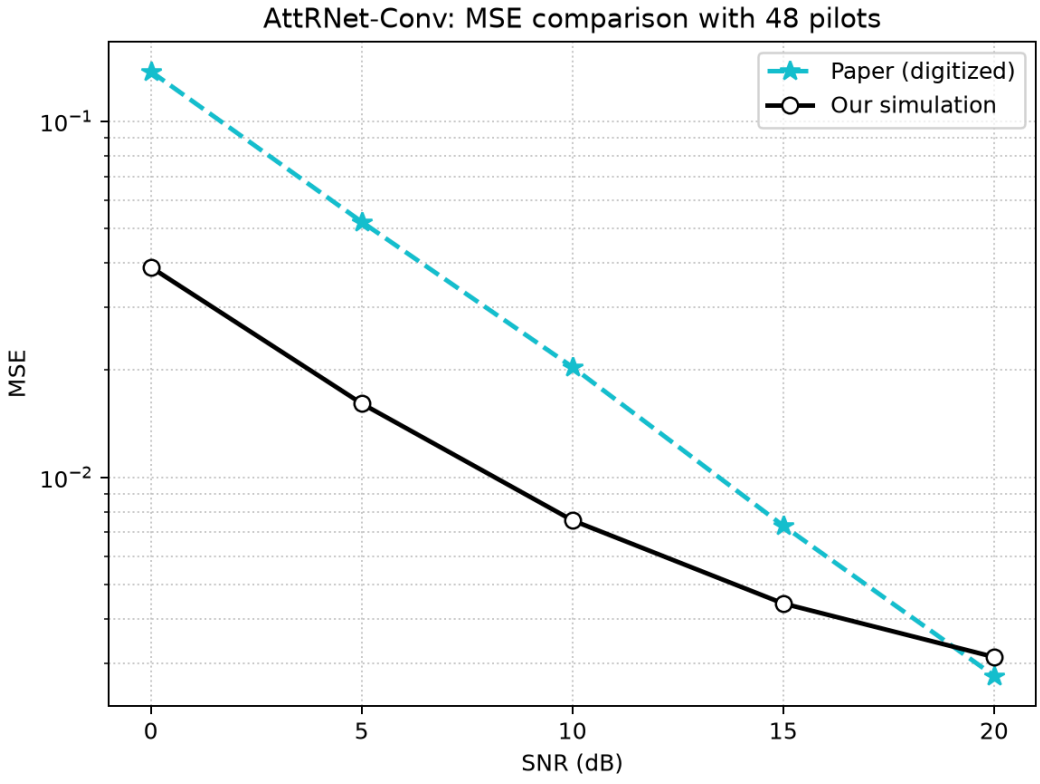


Table 4. AttRNet-Conv with 48 Pilots

SNR (dB)	Paper MSE (approx.)	Reproduced MSE	Difference (%)
0	0.13800	0.038917	-71.80
5	0.05220	0.016113	-69.13
10	0.02040	0.007568	-62.90
15	0.00731	0.004420	-39.53
20	0.00277	0.003127	+12.89



At low and medium SNR values, the reproduced MSE values are lower than the approximate reference values for both architectures and pilot configurations. However, at 20 dB, the reproduced MSE is higher than the reference values for all four configurations. The magnitude of this discrepancy varies by model and pilot count.

4.3 Interpretation

The observed reduction in MSE with increasing SNR is qualitatively consistent with expected channel-estimation behavior. The absolute MSE mismatch is be related to differences in:

- Channel normalization
- Noise power and SNR definition
- Pilot placement
- Channel-object configuration.

5. Limitations

- The original authors' code and dataset are not available to this project, so the dataset and training environment are independently reconstructed.
- Some details including pilot coordinates, SNR mixture and MATLAB channel-object options are not fully specified in the paper·
- The paper comparison uses values digitized from figures and therefore contains reading uncertainty.
- Conventional and deep learning baselines have not yet been evaluated on the same locally generated test dataset.

6. Conclusion

A custom simulation and deep learning pipeline has been established for studying AttRNet-based channel estimation in a conventional EPA-OFDM system. Initial experiments for AttRNet-FC with 24 pilots and AttRNet-Conv with 48 pilots show decreasing MSE with increasing SNR.

The current work should be regarded as an implementation and preliminary replication stage rather than a fully validated reproduction of all reported paper results.

The immediate priority is to validate the simulation and evaluation assumptions and complete the controlled experiments. This will provide a reliable baseline for the subsequent investigation of pilot-efficient channel estimation and the broader proposed channel scenarios.

References

[1] W. Gao, W. Zhang, L. Liu, and M. Yang, “Deep residual learning with attention mechanism for OFDM channel estimation,” IEEE Wireless Communications Letters, vol. 14, no. 2, pp. 250–254, Feb. 2025.

[2] Project repository, MTP_Attnet, https://github.com/priyanshu2393/MTP_Attnet.